PROJECT TECHNICAL SYSTEM BLUEPRINT

Prepared For:	tecsoull
Project Focus:	E-Commerce
Development Stack:	Recommended Architect Choice
Timeline Scheme:	Fast
Total Scope Cost:	■117,760
Generated On:	2026-07-15

1. Executive & Business Summary

Technical blueprint outlining the software architecture and system components for the 'E-Commerce' solution tailored for tecsoull.

2. Recommended System Architecture

Layered MVC Architecture. Front-end client communicates with a secure server-side API Gateway that interacts with internal services and a central transaction database.

3. Product Tech Stack Details

Frontend: React.js (SaaS UI template), Backend: Python Flask / Node.js, Database: PostgreSQL / SQLite, Cloud: Docker containerization, AWS Hosting support.

4. Database Schema Design

Users Table (id, email, password_hash, company, created_at)
Projects Table (id, user_id, name, budget, status, created_at)
Quotations Table (id, project_id, details_json, created_at)
Proposals Table (id, project_id, scope, created_at)

5. User Roles & Permission Matrix

Administrator: Full dashboard control, CRUD capabilities on users/leads, analytics visibility.
Client: Settings updates, file uploads access, download quotation & proposals.
Guest: Basic consultation chat session operations.

6. Screen Interface Flow

1. Authentication (Login & Signup)
2. Customer Portal Dashboard (Leads, Roadmap, Files list)
3. Consultation Screen (Chat interface, cards selectors)
4. Admin CRM Dashboard (Leads tracking table, statistics grid)

7. REST API Integration Endpoints

1. GET /api/start - Initiate chat session
2. POST /api/chat - Submit message/card
3. POST /api/upload - Upload requirement documents
4. GET /api/files - Fetch user documents
5. GET /api/dashboard/summary - Load stats & user data

8. Security & Encryption Standards

PBKDF2 Password hashing algorithm, secure HTTP-only sessions, CSRF token validation, SQL injection prevention through parameterized queries, XSS sanitization.

9. Deployment Pipeline & Infrastructure

Containerization using Docker, Automated pipeline using GitHub Actions, deployment onto AWS EC2 instance with SSL encryption via Nginx reverse proxy.

10. Maintenance & Quality Assurance

Unit testing using Pytest, manual UI verification, compliance debugging support under complimentary 90-day post-launch warranty.

11. Future Expansion Roadmap

Phase 2 scale upgrades including PostgreSQL replication, Redis query caching layer, and integration of specialized AI chatbot agents.